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NOVEL REAL-TIME MONITORING OF HEALTH RISK BEHAVIORS: A CASE FOR PRIVACY-CONSCIOUS WEARABLE CAMERAS

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Numerous health-risk behaviors, including overeating and smoking, are linked to higher morbidity and mortality rates. Preventing these behaviors is essential for disease prevention, necessitating an understanding of, and potential alterations to, individuals' oral consumption patterns. In behavioral medicine, wearable cameras have emerged as a transformative tool, offering valuable visual insights to various end users, such as clinical teams, health coaches, and dietitians. However, concerns about bystander privacy have hindered the widespread adoption of such cameras in daily life, and sensors have been known to confound smoking and eating events, impacting detection accuracy. To address this issue, we developed a novel wearable device called HabitSense, to automatically detect hand-to-mouth gestures, distinguish between eating and smoking events, and record videos focused solely on the wearer and nearby objects while anonymizing distant objects and non-consented bystanders. In this study, we evaluate the acceptability and accuracy of real-time detection in real-world settings, specifically among individuals who smoke and those with obesity.

We designed and developed a neck-worn camera with two primary sensing components: a thermal sensor and an RGB sensor, further complemented by an inertial measurement unit for tracking motion. To thoroughly evaluate its performance, we conducted a 7-day study involving 15 participants who were instructed to wear the device continuously during waking hours, resulting in 800 hours of real-world video data. To enhance the recognition of eating and smoking gestures, we employed advanced transformer-based machine learning architecture for training our models, utilizing the insights gained from our dataset. Trained labelers labeled each hand-to-mouth gesture to enable evaluation of our models.

In our dataset, we collected a significant total of 118,267 hand-to-mouth gestures, comprising approximately 3,065 instances of smoking gestures and approximately 4,598 instances of eating gestures. The outcomes obtained through the training of models on this dataset strongly emphasize the effectiveness of the HabitSense system, as it accurately detected 81% of eating and 88.2% of smoking events. Furthermore, our real-time machine learning model achieved an F1-Score of 0.73 for detecting smoking sessions, employing lightweight and efficient machine learning algorithms.

This advancement in the acceptability and accuracy of wearable cameras holds promise for integrating wearable cameras into population-based settings, improving our understanding and management of health-related behaviors. Moreover, its potential for future enhancements, including just-in-time interventions based on detected activities, highlights its versatility and potential impact on behavioral medicine.

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A MULTIFACETED ANALYSIS OF OVEREATING IN ADULTS WITH OBESITY: ECOLOGICAL MOMENTARY STUDY

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Obesity, a prevailing global health concern, primarily results from overeating, or excessive calorie intake, shaped by context such as psychological and environmental factors. Prior research focused on average effects of these factors among different individuals (referred to as between-subject [BS] effects), while neglecting within-subject [WS] effects (e.g., meal location) and how consistent, or variable, these effects are. This understanding is critical for recommending targeted, personalized interventions to address overeating. In our study, we used a mixed-effects location scale model to account for BS and WS in overeating analysis.

In a 14-day free-living observational study, 47 adults with obesity (BMI ≥ 30 kg/m^2) recorded 2004 meal episodes. Participants donned 3 wearable devices (e.g., wearable thermal camera) for visual meal verification and provided 24-hour dietitian-administered food recalls. Participants also completed Ecological Momentary Assessments (EMA) on affect, stress, hunger, and meal context. To identify factors contributing to overeating, we built a two-level mixed-effects location scale model to EMA data that controlled for both between-subject (BS) and within-subject variability (WS). Our outcome variable was caloric intake per meal estimated by 24-hour diet recalls.

We identified six BS factors (i.e., stress, perception of overeating, restaurant food, later meals, pleasure-seeking meal) and ten WS factors (i.e., biological hunger, perceived overeating, uncontrolled eating, social eating, restaurant food, snacks) that significantly associated with caloric consumption. Notably, participants who were more stressed, on average, consumed more calories (0.74; $p = 0.002$) with high consistency (-0.7; $p = 0.048$) between individuals. Occasion-level stress led to higher caloric consumption variability (1.68; $p = 0.02$) relative to individual averages.

These significant associations highlighted from our multifaceted, free-living observational research offer valuable insights for shaping personalized intervention strategies (e.g., mindfulness-based interventions to reduce stress before eating; just-in-time [JIT] intervention elements to the participant's current emotional status in real time) to combat this obesity crisis.

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